Rethinking Heterogeneous Graph Representation Learning via Relational Subset Selection

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ABSTRACT

Long-range information propagation has been extensively studied in homogeneous graphs. However, directly extending deeper propagation to heterogeneous graphs often leads to severe noise accumulation and prohibitive computation due to the exponential growth of candidate relations. In this work, we argue that the fundamental challenge of heterogeneous graph neural networks (HGNNs) lies not in insufficient propagation depth, but in identifying informative relations from an exponentially large set of meta-paths.

Motivated by this insight, we reformulate heterogeneous representation learning as a relational subset selection problem, shifting the focus from neighborhood expansion to selecting a compact set of informative relations. From this perspective, we propose a progressive meta-path search framework that incrementally reduces the candidate space while preserving useful long-range dependencies. Furthermore, we introduce a progressive single meta-path search strategy with leave-one-out evaluation, which provides a theoretically motivated approximation of marginal contribution and transforms combinatorial subset optimization into an efficient contribution estimation process with hop-independent complexity.

Extensive experiments on multiple heterogeneous benchmarks and the large-scale OGBN-MAG dataset demonstrate that the proposed methods consistently outperform state-of-the-art HGNNs. Moreover, the discovered meta-paths generalize across different architectures, indicating that the framework captures dataset-level relational structures rather than model-specific patterns.

1. Introduction

Heterogeneous graphs (HGs) are widely used to model complex systems with multiple types of entities and relations. For example, the large-scale academic network `OGBN-MAG` contains node types such as Paper (P), Author (A), Institution (I), and Field (F), together with diverse relations including Author $\xrightarrow{\text{writes}}$ Paper, Paper $\xrightarrow{\text{cites}}$ Paper, Author $\xrightarrow{\text{is affiliated with}}$ Institution, and Paper $\xrightarrow{\text{has topic of}}$ Field. These heterogeneous relations can be composed into higher-order semantic structures, commonly known as meta-paths [1]. For instance, APA represents co-authorship, while longer paths such as PFPAPFP encode long-range semantic dependency.

This paper is an extended and improved version of our previous conference paper [2]. Compared to the conference version, this work introduces a progressive single meta-path search strategy with marginal contribution estimation, and provides a more comprehensive formulation and evaluation.

Rethinking long-range dependency in HGs. Long-range dependency is crucial for graph representation learning. In homogeneous graphs, deeper message passing or transformer-style architectures can gradually enlarge receptive fields and improve access to distant information [3, 4, 5, 6]. However, heterogeneous graphs fundamentally differ from homogeneous ones: their relations are semantically diverse, task-dependent, and often highly uneven in usefulness. As a result, expanding neighborhoods does not necessarily improve representation quality. For example, predicting a paper's field may benefit from related papers and authors, but may be distracted by excessive aggregation over weakly relevant institutions or fields.

Therefore, the fundamental challenge is not how far information can propagate, but which relational paths should be propagated.

Limitations of existing HGNN paradigms. Existing heterogeneous graph neural networks (HGNNs) mainly follow three lines of research. Meta-path-free methods [7, 8, 9, 10] stack layers to aggregate l-hop neighborhoods, whose receptive fields grow rapidly with depth and often introduce noisy heterogeneous signals. Transformer-based approaches [11, 6] further enlarge receptive fields through global attention, but suffer from high computational cost and remain difficult to scale to large graphs [12]. Metapath-based methods [13, 14, 15, 16] restrict aggregation to predefined paths, but capturing long-range dependency still requires increasing the maximum hop l, which again leads to an exponential number of candidate meta-paths.

Despite their differences, these paradigms share a common assumption: better heterogeneous representation learning can be achieved by aggregating over more reachable relations. However, this strategy does not explicitly address a more critical question: which relations are actually informative for the downstream task? When useful relations are sparse, enlarging the receptive field inevitably mixes informative and irrelevant signals, leading to a fundamental scalability–noise trade-off.

A subset-selection perspective. Our empirical observation is that heterogeneous tasks are often dominated by only a small subset of meta-paths, while many others are redundant

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or even harmful. This suggests that the bottleneck of long-range heterogeneous learning is not insufficient propagation depth itself, but the lack of effective relation selection.

From this perspective, heterogeneous representation learning can be formulated as a relational subset selection problem, where the goal is to identify a compact set of informative meta-paths from an exponentially large candidate space.

More specifically, let $\mathcal{P}$ denote the full set of target-related meta-paths. Conventional HGNNs implicitly learn representations from all available relations, i.e., $f(\mathcal{G}, \mathcal{P})$. In contrast, our goal is to learn from a compact subset $S \subset \mathcal{P}$ with $|S| = M \ll |\mathcal{P}|$, namely $f(\mathcal{G}, S)$. This formulation shifts the focus from how to aggregate heterogeneous information to what relational structures should be retained.

Our approach. Motivated by this perspective, we propose Long-range Meta-path Search through Progressive Sampling (LMSPS) and its efficient variant Long-range Progressive Single Meta-path Search (LPSMS). Rather than expanding receptive fields or searching for model architectures, our framework directly estimates the utility of candidate meta-paths and performs representation learning only on a selected subset.

LMSPS progressively shrinks the candidate space through stochastic sampling and evaluates sampled subsets with a validation-driven strategy. LPSMS further improves efficiency by transforming subset optimization into marginal contribution estimation via leave-one-out evaluation. This converts combinatorial subset search into a tractable contribution estimation process with hop-independent complexity.

By decoupling relation selection from representation learning, our framework enables effective use of long-range dependencies without suffering from the exponential growth of candidate paths. Moreover, because the search stage is largely independent of the downstream HGNN architecture, the discovered meta-paths can generalize across different backbones, indicating that they capture dataset-level relational structures rather than model-specific behavior.
Our main contributions are summarized as follows:

- We present a subset-selection perspective for long-range heterogeneous representation learning, showing that the central challenge is relation selection rather than propagation depth.
- We propose LMSPS, a progressive sampling framework that efficiently identifies informative meta-path subsets from exponentially large candidate spaces.
- We introduce LPSMS, which estimates marginal contributions of individual meta-paths and improves both search efficiency and stability.
- Extensive experiments on multiple heterogeneous graph benchmarks demonstrate superior accuracy, scalability, and cross-model generalization of the proposed framework.

2. Preliminaries

Heterogeneous Graph. [17] A heterogeneous graph is defined as $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathcal{T}, \mathcal{R}, f_{\mathcal{T}}, f_{\mathcal{R}}\}$ with $|\mathcal{T}+\mathcal{R}| > 2$, where $\mathcal{V}$ and $\mathcal{E}$ denote nodes and edges, $\mathcal{T}$ and $\mathcal{R}$ denote node and edge types, and $f_{\mathcal{T}} : \mathcal{V} \to \mathcal{T}$, $f_{\mathcal{R}} : \mathcal{E} \to \mathcal{R}$ assign semantic types to nodes and relations.

Meta-path. [1] A meta-path P is a sequence of relations

$$P = c_1 \xleftarrow{r_{12}} c_2 \cdots c_{l-1} \xleftarrow{r_{(l-1)l}} c_l,$$

where $c_i \in \mathcal{T}$ and $r_{ij} \in \mathcal{R}$. Each meta-path corresponds to a set of path instances in $\mathcal{G}$ and defines a semantic connectivity pattern. Aggregation along different meta-paths extracts heterogeneous information under distinct semantics.

Long-range representation learning. Let $\mathcal{P}$ denote the set of all candidate meta-paths related to a target node type. Traditional HGNNs aggregate representations over all reachable neighbors, which implicitly corresponds to using the full set $\mathcal{P}$. However, heterogeneous relations are often noisy or redundant, and only a subset contributes positively to downstream prediction.

Path subset selection problem. Instead of aggregating all candidate paths, heterogeneous representation learning can be formulated as selecting an informative subset

$$\min_{S \subseteq \mathcal{P}, |S|=M} \mathcal{L}\big(f(\mathcal{G}, S)\big),$$

where S is a compact meta-path set and $M \ll |\mathcal{P}|$. This formulation shifts the goal from neighborhood expansion to identifying informative relational structures. The difficulty arises because $|\mathcal{P}|$ grows exponentially with hop length, making direct search intractable.

3. Related Works

Neighborhood expansion in HGNNs. Most heterogeneous graph neural networks (HGNNs) follow a neighborhood expansion paradigm, where representation learning is achieved by aggregating information from multi-hop neighbors. Meta-path-free models [7, 8, 9, 10] stack multiple layers to aggregate l-hop neighbors, implicitly assuming that larger receptive fields improve representation quality. Transformer-based approaches [11] further enlarge receptive fields via global attention, but suffer from quadratic complexity and scalability limitations on large graphs. Metapath-based models [13, 14, 15, 16] restrict aggregation along predefined paths, yet still rely on enumerating all candidate meta-paths within a given hop.

Despite their differences, these approaches share a common assumption: long-range dependency should be captured by expanding receptive fields. However, in heterogeneous graphs, relations are semantically diverse and often task-dependent. Blindly expanding neighborhoods inevitably mixes informative and irrelevant signals, leading to a fundamental scalability–noise trade-off. More importantly, these methods implicitly assume that aggregating more relations improves representation quality, while our empirical

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findings suggest that many relations are redundant or even harmful. Consequently, these approaches do not explicitly address a more critical question: *which relations should be used for representation learning*.

Implicit meta-path learning. Several works attempt to implicitly learn useful relational structures through attention mechanisms or adaptive aggregation. GTN [18] and HGT [9] can discover meta-path-like structures, but the learned relations are not explicitly optimized for final prediction. Methods such as HAN [13], HPN [15], and MEGNN [19] analyze path importance only as an auxiliary signal rather than a primary objective. GraphMSE [20] evaluates discovered meta-paths but still observes that using all relations often yields better performance, suggesting that implicit weighting is insufficient to filter out noisy or redundant relations.

In general, these approaches treat relation discovery as a by-product of representation learning, which limits their ability to explicitly model relational utility and perform effective selection. As a result, they lack a mechanism to identify a compact subset of informative relations under an exponentially large candidate space.

Meta-structure search methods. Neural architecture search (NAS) has been introduced to automatically discover heterogeneous structures. Methods such as GEMS [21], DiffMG [22], and PMMM [23] search meta-graphs, while RL-HGNN [24] and RMS-HRec [25] search meta-paths using reinforcement learning.

These approaches formulate structure discovery as an architecture optimization problem, where the objective is to identify an optimal relational structure for a specific model. However, their optimization cost grows with the size of the combinatorial search space, making them difficult to scale to long-range heterogeneous dependencies. Moreover, they typically aim to find a single best structure, rather than identifying a compact subset of informative relations among exponentially many candidates. This limits their ability to capture the sparse utility pattern observed in heterogeneous graphs.

Our perspective. In contrast to the above paradigms, we reformulate heterogeneous representation learning as a relational subset selection problem. Rather than expanding neighborhoods or searching model structures, our approach directly estimates the utility of candidate relations and selects a compact informative subset. This perspective shifts the focus from *how to aggregate* to *what to aggregate*, and explicitly models the sparse utility structure of heterogeneous relations, enabling scalable and noise-aware modeling of long-range dependencies.

4. Empirical Motivation

Long-range dependency does not imply full aggregation. Long-range meta-paths provide complementary relational information beyond immediate neighbors. For example, the meta-path MDMDMK in IMDB connects movies through co-directors and shared keywords, allowing a movie without keywords to indirectly obtain semantic cues from distant nodes. This shows that long-range dependency is beneficial in heterogeneous graphs. However, heterogeneous relations are semantically diverse, and not all long-range connections are helpful. Therefore, the key challenge is not how far information propagates, but which relational paths should be used.

Empirical evidence. To examine the contribution of different meta-paths, we analyze SeHGNN [16] on DBLP and ACM from the HGB benchmark [10]. We restrict the maximum hop to $l = 2$ to enumerate all candidate meta-paths and evaluate their individual and joint contributions.

On DBLP, removing most meta-paths causes negligible performance change, while removing APV leads to a significant drop (Figure 1(a)). Moreover, using APV alone achieves nearly identical performance to using all paths (Figure 1(b)), indicating that prediction performance is dominated by a small number of meta-paths. On ACM, removing certain meta-paths even improves performance (Figure 1(c)), showing that some relations introduce noise.

These results suggest that heterogeneous relations exhibit a sparse utility pattern: only a small subset of meta-paths is informative, while many others are redundant or harmful.

Problem implication. Existing HGNNs aggregate information over all reachable nodes or predefined paths. When informative relations are sparse, expanding receptive fields inevitably mixes useful and noisy signals, which explains why most models restrict hop size and why attention alone cannot remove irrelevant relations. Let $\mathcal{P}$ denote all candidate meta-paths. Although $|\mathcal{P}|$ grows exponentially with hop length, the effective set follows a Pareto-like distribution [26, 27], where only a small fraction contributes to prediction.

Therefore, long-range heterogeneous representation learning can be viewed as identifying an informative subset of meta-paths from an exponentially large candidate space. The remainder of this paper focuses on solving this subset selection problem.

5. Methodology

Following the formulation in the previous sections, our goal is to approximate the solution of the meta-path subset selection problem, i.e., reducing the exponentially large candidate meta-path space to a compact subset that is effective for the given dataset and task. Instead of aggregating information from all meta-paths, we first identify informative paths and then perform representation learning using only the selected subset.

This design brings two advantages. First, restricting aggregation to effective meta-paths reduces computational and memory cost while preserving useful heterogeneous information. Second, each target node aggregates neighbors only along instances of the selected meta-paths. Since informative meta-paths occupy only a small portion of the candidate space and nodes have different neighborhood structures, different nodes aggregate different relational contexts under the

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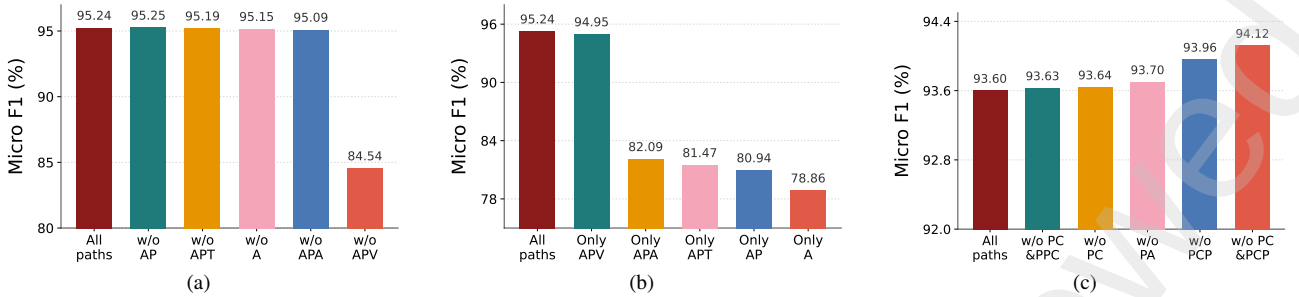


Figure 1: Analysis of the importance of different meta-paths. (a) illustrates the results after removing a single meta-path on DBLP; (b) shows the performance of utilizing a single meta-path on DBLP; (c) illustrates the performance after removing a part of meta-paths on ACM.

constraint of selected paths, thereby avoiding unnecessary mixing of heterogeneous signals.

The remaining challenge is how to identify effective meta-paths, especially long-range ones. Searching long-range meta-paths faces two main difficulties: the exponential growth of candidates and the existence of noisy relations. We first design a progressive sampling algorithm to address the exponential growth issue. We then develop a progressive single meta-path search algorithm to further improve search efficiency and alleviate under-exploration. Figure 2 illustrates the overall framework of LMSPS and LPSMS, consisting of a super-net for subset discovery and a target-net for representation learning.

The super-net aims to estimate meta-path utility independent of downstream prediction modules, so that the search results are not biased by specific architectures. Based on this consideration, we adopt a simple MLP-based architecture rather than a transformer-based one for meta-path search, since it introduces fewer modeling assumptions [28, 29, 30]. The super-net contains five components: neighbor aggregation, feature projection, progressive sampling search, sampling evaluation, and MLP.

Let $\mathbb{P} = \{P_1, \cdots, P_k, \cdots, P_K\}$ be the initial search space containing all K target-related meta-paths, $\boldsymbol{X}^{c_i}$ be the raw feature matrix of nodes belonging to type c_i, and $\hat{\boldsymbol{A}}_{c_i,c_{i+1}}$ be the row-normalized adjacency matrix between node type c_i and c_{i+1}. The neighbor aggregation block follows SeHGNN [16], which performs a one-time message passing pre-processing to transform the heterogeneous graph into path-specific tensors for target nodes. Specifically, it multiplies adjacency matrices to compute the contribution weights of meta-path-based neighbors. As shown in Figure 2, the aggregation of an l-hop meta-path $P_k = c_0 c_1 c_2 \ldots c_l$ is defined as:

$$\boldsymbol{X}_k = \begin{cases} \boldsymbol{X}^{c_0} & l = 0 \\ \hat{\boldsymbol{A}}_{c_0,c_1} \hat{\boldsymbol{A}}_{c_1,c_2} \cdots \hat{\boldsymbol{A}}_{c_{l-1},c_l} \boldsymbol{X}^{c_l} & l > 0 \end{cases}, \tag{1}$$

where $\boldsymbol{X}_k$ denotes the feature matrix corresponding to meta-path P_k, and c_0 is the target node type. The feature projection block then maps all meta-path features into a shared latent space: $\boldsymbol{X}'_k = \mathrm{MLP}_k(\boldsymbol{X}_k)$.

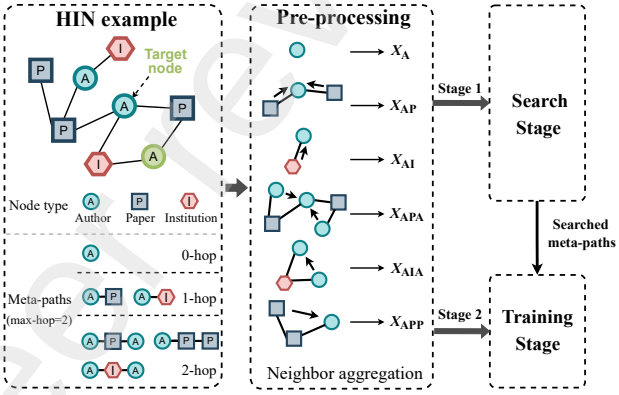


Figure 2: Overall framework of LMSPS and LPSMS. The framework decouples relational subset selection from representation learning. In the search stage, informative meta-paths are identified from K candidates via progressive sampling and evaluation, resulting in a compact subset of size M. In the training stage, representation learning is performed using only the selected meta-paths. Aggregation over 0-, 1-, and 2-hop meta-paths is illustrated for clarity, while feature projection updates are omitted.

5.1. The Search Algorithm

The search stage aims to identify the most effective meta-path subset from all candidates, which is computationally challenging. For example, in OGBN-MAG, the number of target-related meta-paths is $K = 226$ (under maximum hop 6), and the goal is to select $M = 30$ paths. Since meta-paths may be noisy or redundant to each other, selecting the top-M paths independently does not guarantee the optimal subset. The number of possible subsets is $C_{226}^{30} \approx 10^{37}$, making direct search infeasible for RL-based [24, 25] or evolutionary algorithms [31, 32].

To address this issue, we first apply a search procedure to shrink the candidate space from 226 to 60, and then employ a sampling evaluation strategy to identify the subset with the lowest validation loss, which is more reliable than relying solely on architecture parameters [33].

Additionally, restricting the number of selected meta-paths to a constant M stabilizes the size of aggregated neighborhoods across different hop lengths. By avoiding aggregation over exponentially growing relational structures,

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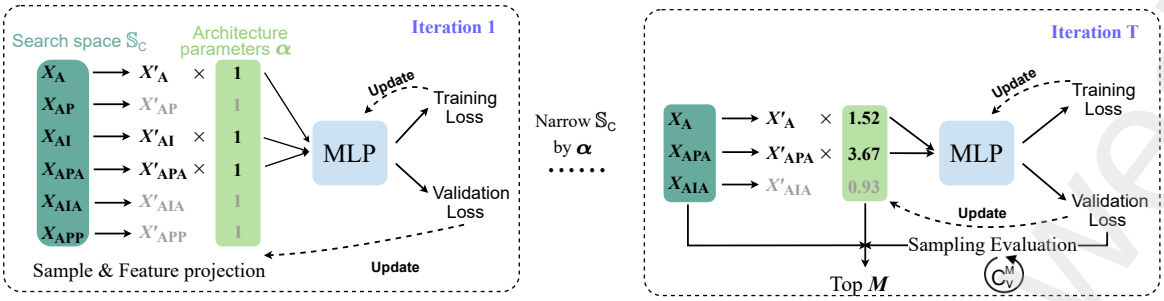


Figure 3: Workflow of LMSPS: progressive exploration of multi-path relation combinations.

the model focuses on informative relational channels when learning representations.

5.2. Progressive Sampling Search

Algorithm 1 Progressive Sampling Search of LMSPS

Input: meta-path space $\mathbb{P} = \{P_1, \cdots, P_K\}$; target subset size M; search iterations T; evaluation trials E

Parameters: network weights $\boldsymbol{\omega}$; architecture parameters $\boldsymbol{\alpha} = \{\alpha_1, \cdots, \alpha_K\}$

Output: selected meta-path subset $\mathbb{S}_M$

// Pre-compute path features
for each $P_k \in \mathbb{P}$ **do**
  compute $\boldsymbol{X}_k$ using Eq. (1)
end for
// Progressive search stage
for $t = 1 \to T$ **do**
  Compute path strengths $\{q_k\}$ using Eq. (2)
  Construct dynamic candidate space $\mathbb{S}_C$ using Eq. (4)
  Sample a subset $\mathbb{S} \sim \text{UniformSample}(\mathbb{S}_C, M)$
  Update network weights:

$$\boldsymbol{\omega} \leftarrow \boldsymbol{\omega} - \eta_\omega \nabla_{\boldsymbol{\omega}} \mathcal{L}_{train}(\mathbb{S})$$

  Update architecture parameters:

$$\boldsymbol{\alpha} \leftarrow \boldsymbol{\alpha} - \eta_\alpha \nabla_{\boldsymbol{\alpha}} \mathcal{L}_{val}(\mathbb{S})$$

end for
// Subset evaluation stage
Fix compact search space $\mathbb{S}_V$
for $e = 1 \to E$ **do**
  Sample subset $\bar{\mathbb{S}} \sim \text{DiscreteSample}(\mathbb{S}_V, M)$
  Compute validation loss $L_e = \mathcal{L}_{val}(\bar{\mathbb{S}})$
end for
$\mathbb{S}_M \leftarrow \arg\min_{\bar{\mathbb{S}}} L_e$
return $\mathbb{S}_M$

Since the search space contains all target-related meta-paths, its size grows exponentially with hop length, making direct subset optimization infeasible. To address this issue, LMSPS uses a progressive sampling strategy that iteratively estimates path utility and shrinks the candidate space.

We assign each meta-path an architecture parameter $\boldsymbol{\alpha} = \{\alpha_1, \cdots, \alpha_K\} \in \mathbb{R}^K$. The relative contribution of each path is estimated using Gumbel-softmax [34, 35]:

$$q_k = \frac{\exp\left[\left(\alpha_k + u_k\right)/\tau\right]}{\sum_{j=1}^{K} \exp\left[\left(\alpha_j + u_j\right)/\tau\right]}, \tag{2}$$

where $u_k = -\log(-\log(U))$ with $U \sim \text{Uniform}(0,1)$ and τ controls the relaxation.

Using the estimated path strength, the candidate space is gradually reduced from K to V. Let $\widetilde{q}_C$ denote the C-th largest element of $\mathbb{Q} = \{q_1, \cdots, q_K\}$. The dynamic search space is defined as

$$\mathbb{S}_C = \{k \mid q_k \geq \widetilde{q}_C\}, \tag{3}$$

$$C = \lceil \lambda(K - V) \rceil + V, \tag{4}$$

where $\lambda \in [0,1]$ decreases during training.

This progressive pruning can be viewed as a coarse-to-fine estimation process over relational utility. In early iterations, the estimated strengths $\{q_k\}$ are noisy but provide a useful signal of relative importance. As training proceeds, the ranking induced by $\{q_k\}$ becomes increasingly aligned with the underlying utility of meta-paths. Under this view, restricting the search to $\mathbb{S}_C$ preserves high-utility candidates with high probability, while eliminating a large number of low-utility relations from further consideration. This converts the exponentially large search space into a compact yet informative candidate set.

At each iteration, M meta-paths are sampled from $\mathbb{S}_C$ and only their parameters are updated. Thus the search cost depends on M rather than K. The forward propagation becomes:

$$\boldsymbol{Z} = \text{MLP}\Big(\sum_{k \in \mathbb{S}} q'_k \cdot \text{MLP}_k(\boldsymbol{X}_k)\Big), \tag{5}$$

$$\text{where} \quad \mathbb{S} = \text{UniformSample}(\mathbb{S}_C, M). \tag{6}$$

Here q'_k is the normalized strength among activated paths.

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The parameter learning forms a bilevel optimization problem [36, 37, 38]:

$$\min_{\boldsymbol{\alpha}} \mathcal{L}_{val}(\mathbb{S}, \boldsymbol{\omega}^*(\boldsymbol{\alpha}), \boldsymbol{\alpha}), \tag{7}$$

$$\text{s.t. } \boldsymbol{\omega}^*(\boldsymbol{\alpha}) = \arg\min_{\boldsymbol{\omega}} \mathcal{L}_{train}(\mathbb{S}, \boldsymbol{\omega}, \boldsymbol{\alpha}). \tag{8}$$

We solve it via alternating optimization: fixing $\boldsymbol{\alpha}$ when updating $\boldsymbol{\omega}$ on the training set, and fixing $\boldsymbol{\omega}$ when updating $\boldsymbol{\alpha}$ on the validation set.

The progressive sampling mechanism produces a compact candidate set guided by the dataset and task, while the stochastic sampling in each iteration reduces coupling among meta-paths.

Theoretical justification (Approximate subset selection). The progressive sampling strategy can be viewed as an approximate optimization procedure for relational subset selection. Specifically, by iteratively updating the architecture parameters $\boldsymbol{\alpha}$ using validation loss, the induced path strength $\{q_k\}$ serves as a surrogate estimate of the marginal utility of each meta-path. Under the weak interaction assumption (Assumption 1), this iterative pruning process approximates selecting a high-utility subset from an exponentially large candidate space, while avoiding explicit combinatorial optimization.

After progressive pruning, the candidate space is reduced from K to V. Because meta-path contributions are not independent, ranking paths individually does not guarantee an optimal subset. Therefore we evaluate subsets directly through stochastic subset evaluation.

Using the final path strength as sampling probability, we repeatedly sample M meta-paths from $\mathbb{S}_V$ and select the subset with lowest validation loss:

$$\bar{\mathbb{S}} = \text{DiscreteSample}(\mathbb{S}_V, M, \bar{\mathbb{Q}}) \tag{9}$$

The final subset is

$$\mathbb{S}_M = \arg\min_{\bar{\mathbb{S}}} \mathcal{L}_{val}(\bar{\mathbb{S}}, \boldsymbol{\omega}^*, \boldsymbol{\alpha}^*)$$

and the target-net performs representation learning as

$$\hat{\boldsymbol{Z}} = \text{MLP}\Big(\Big\Vert_{k \in \mathbb{S}_M} \text{MLP}_k(\boldsymbol{X}_k) \Big). \tag{10}$$

Here $\Vert$ denotes concatenation.

5.3. Progressive Single Meta-path Search

The progressive sampling search in LMSPS identifies useful meta-path subsets by repeatedly evaluating randomly sampled path sets. Although effective, the optimization cost still depends on the subset size M and requires many stochastic evaluations to approximate the optimal combination. This motivates a more direct strategy: instead of estimating subset quality through repeated sampling, we estimate the *marginal contribution* of each meta-path.

Algorithm 2 Progressive Single Meta-path Search of LPSMS

Input: meta-path space $\mathbb{P} = \{P_1, \cdots, P_K\}$; target subset size M; search iterations T

Parameters: network weights $\boldsymbol{\omega}$; architecture parameters $\boldsymbol{\alpha} = \{\alpha_1, \cdots, \alpha_K\}$

Output: selected meta-path subset $\mathbb{S}_M$

// Pre-compute path features
for each $P_k \in \mathbb{P}$ **do**
  compute $\boldsymbol{X}_k$ using Eq. (1)
end for
// Progressive single-path optimization
for $t = 1 \to T$ **do**
  Compute path strengths $\{q_k\}$ using Eq. (2)
  Construct dynamic candidate space $\mathbb{S}_C$ using Eq. (4)
  Sample one path $k \sim \text{UniformSample}(\mathbb{S}_C, 1)$
  Update network weights:

$$\boldsymbol{\omega} \leftarrow \boldsymbol{\omega} - \eta_\omega \nabla_{\boldsymbol{\omega}} \mathcal{L}_{train}(k)$$

  Update architecture parameters:

$$\boldsymbol{\alpha} \leftarrow \boldsymbol{\alpha} - \eta_\alpha \nabla_{\boldsymbol{\alpha}} \mathcal{L}_{val}(k)$$

end for
// Marginal contribution estimation
Obtain compact candidate set $\mathbb{S}_V$
for each $P_i \in \mathbb{S}_V$ **do**
  Evaluate validation loss after removing P_i:

$$\Delta_i = \mathcal{L}_{val}(\mathbb{S}_V \setminus \{P_i\}) - \mathcal{L}_{val}(\mathbb{S}_V)$$

end for
$\mathbb{S}_M \leftarrow$ Top-M paths ranked by Δ_i
return $\mathbb{S}_M$

This transformation is critical: it shifts the problem from combinatorial subset optimization to contribution estimation over individual relations. Instead of exploring an exponential number of subsets, we approximate relational utility through marginal effects, which significantly reduces the search complexity while preserving interaction awareness.

Therefore, we propose LPSMS, a progressive single meta-path search algorithm that evaluates paths individually while still considering their contextual interaction. Rather than sampling a subset of paths in each iteration, we activate only one meta-path at a time and learn its utility signal through optimization dynamics. This reduces the search complexity from subset estimation to contribution estimation, effectively approximating a combinatorial optimization problem with a tractable linear evaluation process.

Single-path optimization. During search, one candidate meta-path is sampled from the dynamic search space $\mathbb{S}_C$

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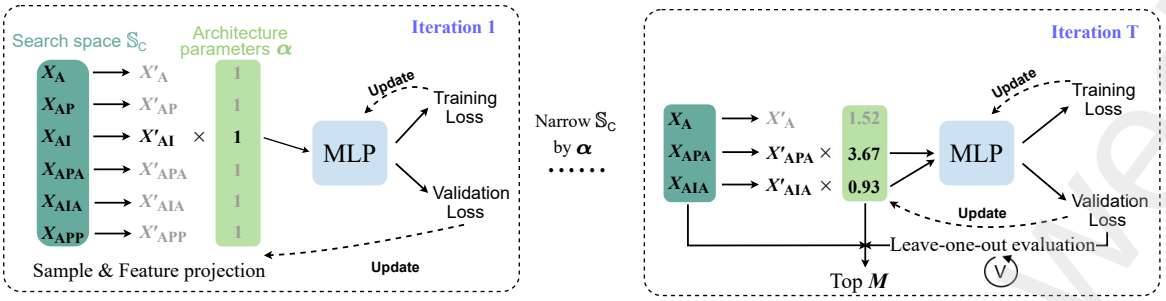


Figure 4: Workflow of LPSMS: single-path progressive exploration with reduced coupling.

per iteration. Only parameters associated with this path are updated:

$$\boldsymbol{Z} = \mathrm{MLP}\Big(q'_k \cdot \mathrm{MLP}_k(\boldsymbol{X}_k)\Big), \quad k \sim \mathrm{UniformSample}(\mathbb{S}_C, 1) \tag{11}$$

By restricting optimization to a single active path at each step, the learning process isolates the contribution signal of individual relations. This avoids interference among multiple paths and provides a clearer estimate of relational utility.

Since each step optimizes a single path, the optimization cost becomes independent of M and scales linearly with K.

Marginal contribution estimation. After progressive pruning reduces the search space from K to V, we estimate the importance of each remaining meta-path using leave-one-out evaluation. Let $\mathbb{S}_M$ be the current candidate set. For each path P_i, we measure the performance change after removing it:

$$\boldsymbol{Z}^{(-i)} = \mathrm{MLP}\Big(\sum_{k \in \mathbb{S}_M \setminus \{i\}} q'_k \cdot \mathrm{MLP}_k(\boldsymbol{X}_k)\Big) \tag{12}$$

The importance score is defined as

$$\Delta_i = \mathcal{L}_{val}(\boldsymbol{Z}^{(-i)}) - \mathcal{L}_{val}(\boldsymbol{Z})$$

A larger Δ_i indicates stronger performance degradation after removal, meaning the meta-path provides more unique information.

This formulation provides a theoretically motivated approximation of marginal utility under the learned representation space, bridging subset selection and contribution estimation. Unlike attention-based weighting, which assigns soft importance scores, our method directly evaluates the necessity of each relation in the presence of others.

By ranking Δ_i, the final subset is obtained as

$$\mathbb{S}_M = \text{Top-}M(\{\Delta_i\})$$

Theoretical justification (Intuition and approximation). To better understand why marginal contribution estimation is effective, we provide an intuitive explanation based on the structure of heterogeneous relational utility.

Let $F(S) = -\mathcal{L}_{val}(S)$ denote the validation utility of a meta-path subset S. Empirically, heterogeneous tasks often exhibit a sparse utility pattern, where only a small number of meta-paths contribute substantially to prediction, while higher-order interactions among most paths are limited.

Under this view, the subset utility can be approximately decomposed as

$$F(S) \approx \sum_{P_i \in S} \phi_i + \epsilon(S),$$

where ϕ_i denotes the first-order utility of meta-path P_i, and $\epsilon(S)$ is a bounded interaction term.

Assumption 1 (Weak Interaction). We assume that the interaction term $\epsilon(S)$ is bounded such that its variation under single-path perturbation is small, i.e.,

$$|\epsilon(S) - \epsilon(S \setminus \{P_i\})| \le \delta,$$

where δ is relatively small compared to the first-order utility ϕ_i. This assumption is consistent with the empirical observations in Figure 1, where only a small subset of meta-paths dominates performance.

Proposition 1 (Marginal Contribution Approximation). Under Assumption 1, the leave-one-out score

$$\Delta_i = \mathcal{L}_{val}(S \setminus \{P_i\}) - \mathcal{L}_{val}(S)$$

provides a first-order approximation to the marginal utility ϕ_i.

Proof (Sketch). By definition,

$$F(S) - F(S \setminus \{P_i\}) \approx \phi_i + \epsilon(S) - \epsilon(S \setminus \{P_i\}).$$

Under the bounded interaction assumption, the difference in interaction terms is limited, and the marginal effect is dominated by ϕ_i. Therefore, Δ_i serves as a first-order estimator of the marginal contribution.

This perspective connects combinatorial subset optimization with marginal contribution estimation: rather than exhaustively evaluating exponentially many subsets, we approximate subset quality by estimating the first-order utility of each relation. From this viewpoint, leave-one-out evaluation provides a practical surrogate for marginal gain estimation.

Although this interpretation does not provide a strict optimality guarantee, it explains why ranking meta-paths

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Table 1
Computational complexity categorized by scalability paradigm. l denotes hop length, r relation types, d average degree, F feature dimension, and M selected meta-paths ($M \ll r^l$).

Paradigm	Scaling Source	Step Complexity	Dependence on l
Neighbor Expansion	Message passing (HAN, Simple-HGN)	$O\big(N(rd)^l F\big)$	Exponential
Meta-path Enumeration	Enumerating all paths ($K = O(r^l)$, SeHGNN)	$O\big(NKF^2\big)$	Exponential
Subset Selection (Ours)	Optimizing sampled path subsets	$O\big(NMF^2\big)$	**Independent**
Marginal Contribution (Ours)	Single-path optimization	$O\big(NF^2\big)$	**Independent**

by Δ_i yields an effective and computationally efficient approximation to relational subset selection in heterogeneous graphs.

Final representation learning. The target network then aggregates only the selected informative paths:

$$\hat{\boldsymbol{Z}} = \mathrm{MLP}\Big(\Big\Vert_{k \in \mathbb{S}_M} \mathrm{MLP}_k(\boldsymbol{X}_k) \Big) \tag{13}$$

Unlike conventional HGNNs, the target-net does not require neighbor attention or semantic attention. All relational selection is completed in the search stage, and the final model becomes a deterministic aggregation over an informative relational basis.

This decoupling of relation selection and representation learning further improves robustness and generalization, as the selected meta-paths capture dataset-level relational patterns rather than model-specific behaviors.

5.4. Time Complexity Analysis

We analyze the computational complexity of LMSPS and LPSMS and compare them with representative HGNNs, including HAN [13], Simple-HGN [10], and SeHGNN [16]. Assume mini-batch training with N target nodes and feature dimension F. Let r denote the number of relation types and d the average degree per relation. The expected number of reachable neighbors within l hops is $(rd)^l$. The results are summarized in Table 1.

Neighbor-expansion models. HAN and Simple-HGN perform message passing during training. Because the receptive field expands exponentially with hop length, their aggregation complexity is

$$O\big(N(rd)^l F\big),$$

which directly depends on neighborhood expansion.

Meta-path enumeration models. SeHGNN avoids runtime neighbor expansion through preprocessing, but requires aggregating all candidate meta-paths. The number of meta-paths satisfies $K = O(r^l)$, leading to

$$O(NKF^2) = O(Nr^l F^2).$$

Subset selection models. Our method activates only a fixed number M of meta-paths, independent of l. Both search and training therefore require

$$O(NMF^2).$$

Marginal contribution search. LPSMS updates a single meta-path per iteration, reducing the search cost to

$$O(NF^2).$$

Summary. Existing HGNNs scale exponentially with hop length due to either neighbor expansion ($(rd)^l$) or path enumeration (r^l). In contrast, both LMSPS and LPSMS have complexity independent of l, making long-range meta-path modeling computationally feasible.

5.5. Comparison with Prior Work

Previous automatic meta-path discovery methods [21, 18, 24] treat the problem as architecture search over relational structures. Their optimization cost grows with the size of the combinatorial structure space, which limits scalability on large heterogeneous graphs.

Our framework instead formulates heterogeneous representation learning as *relational subset selection*. Rather than optimizing model structures, we estimate the contribution of candidate relations and retain only informative ones.

This leads to three conceptual differences.

Scalability. Prior methods scale with the number of candidate structures, while our approach scales with the number of retained structures, which remains constant even when the candidate space grows exponentially.

Long-range modeling. Conventional HGNNs restrict hop size to avoid excessive computation and noise propagation. Our method enables large-hop exploration while controlling noise through selective aggregation.

Transferability. Since the search stage is decoupled from the predictor and uses an MLP with minimal inductive bias, the discovered meta-paths can be reused by other HGNNs (Figure 9), suggesting they capture dataset-level relational patterns rather than model-specific behaviors.

Difference between LMSPS and LPSMS. LMSPS uses subset sampling: it repeatedly samples multiple meta-paths and evaluates their combined performance, so the cost depends on subset size M. LPSMS uses single-path evaluation: it evaluates one meta-path at a time and estimates its importance via leave-one-out, reducing the search to marginal contribution estimation and making it more efficient.

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Table 2
Statistics of the datasets used in the experiments.

Dataset	Graph Statistics				Task	
	#Nodes	#Node Types	#Edges	#Edge Types	Target	#Classes
DBLP	26,128	4	239,566	6	author	4
IMDB	21,420	4	86,642	6	movie	5
ACM	10,942	4	547,872	8	paper	3
Freebase	180,098	8	1,057,688	36	book	7
OGBN-MAG	1,939,743	4	21,111,007	4	paper	349
OGBN-MAG-50	1,939,743	4	9,531,403	4	paper	349
OGBN-MAG-20	1,939,743	4	7,975,003	4	paper	349
OGBN-MAG-10	1,939,743	4	6,610,464	4	paper	349
OGBN-MAG-5	1,939,743	4	4,958,941	4	paper	349

6. Experiments

We conduct a comprehensive evaluation of the proposed methods on multiple heterogeneous graph benchmarks, with the goal of assessing their effectiveness, scalability, and robustness.

Our experiments are designed to verify whether modeling heterogeneous representation learning as a relational subset selection problem can improve both predictive performance and structural generalization.

6.1. Experimental Setup

Datasets. We evaluate our method on a diverse set of heterogeneous graphs, including four widely-used benchmarks from HGB [10] (`DBLP`, `IMDB`, `ACM`, `Freebase`) and the large-scale dataset `OGBN-MAG` from OGB [39].

These datasets cover a wide range of structural characteristics, including different numbers of node and relation types, graph densities, and semantic complexities. This diversity allows us to evaluate the robustness of our method across heterogeneous scenarios.

For HGB datasets, we follow the standard transductive setting with splits of 24% training, 6% validation, and 70% testing. For `OGBN-MAG`, we adopt the official temporal split.

To further examine robustness under limited structural information, we construct sparse variants of `OGBN-MAG` by restricting the maximum in-degree per relation type. This enables controlled analysis of performance under varying graph sparsity.

Detailed dataset statistics are summarized in Table 2.

Task. We focus on semi-supervised node classification, which is a standard benchmark for evaluating representation learning in heterogeneous graphs. Each dataset specifies a target node type, and the objective is to predict labels for nodes of this type using both structural and attribute information.

Objective. Following standard practice, we append an MLP classifier to map node representations to class probabilities:

$$\bar{\boldsymbol{Y}} = \mathrm{softmax}(\boldsymbol{Z}\boldsymbol{\omega}_o), \tag{14}$$

and optimize the cross-entropy loss over labeled nodes.

During the search stage, we adopt a bi-level optimization strategy, where training data updates model parameters and validation data guides relation selection. This ensures that the selected meta-paths generalize beyond the training set.

6.2. Baselines and Implementation Details

We compare LMSPS and LPSMS with a comprehensive set of HGNN baselines, covering homogeneous GNNs, relation-aware models, meta-path-based models, and structure search approaches.

The baselines include MLP [40], GraphSAGE [41], RGCN [42], HAN [13], GTN [18], RSHN [7], HetGNN [43], MAGNN [14], HetSANN [8], GCN [44], GAT [45], Simple-HGN [10], HGT [9], SlotGAT [46], DiffMG [22], PMMM [23], HINormer [11], NARS [47], space4HGNN [48], and Se-HGNN [16].

These methods span different modeling paradigms, allowing us to evaluate performance improvements from both architectural and relational perspectives. For large-scale datasets, we include only methods that can run without out-of-memory issues, ensuring fair comparison under realistic computational constraints.

Implementation details. We follow the standard HGB training protocol with early stopping based on validation performance. Each experiment is repeated 5 times for HGB datasets and 10 times for OGBN-MAG, and we report mean performance and standard deviation.

Our model uses $M = 30$ selected meta-paths and a final candidate space size $V = 60$. The maximum hop varies by dataset scale. All experiments are implemented in PyTorch and conducted on a Tesla V100 GPU.

6.3. Overall Performance Analysis

Table 3 summarizes the performance of all methods on both small-scale and large-scale datasets. The results consistently demonstrate the effectiveness of modeling heterogeneous representation learning as a relational subset selection problem.

Relational selection improves predictive performance. Both LMSPS and LPSMS consistently outperform a wide range of HGNN baselines across all datasets. The improvement holds for different graph scales and structural

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Table 3
Performance on small and large datasets. Best is in bold, and the runner-up is underlined.

Method	Small-scale Datasets								Large-scale Dataset
	DBLP		IMDB		ACM		Freebase		OGBN-MAG*
	Macro-F1	Micro-F1	Macro-F1	Micro-F1	Macro-F1	Micro-F1	Macro-F1	Micro-F1	Test Acc.
MLP [40]	-	-	-	-	-	-	-	-	26.92 ± 0.26
GraphSAGE [41]	-	-	-	-	-	-	-	-	46.78 ± 0.67
RGCN [42]	91.52 ± 0.50	92.07 ± 0.50	58.85 ± 0.26	62.05 ± 0.15	91.55 ± 0.74	91.41 ± 0.75	46.78 ± 0.77	58.33 ± 1.57	47.37 ± 0.48
HAN [13]	91.67 ± 0.49	92.05 ± 0.62	57.74 ± 0.96	64.63 ± 0.58	90.89 ± 0.43	90.79 ± 0.43	21.31 ± 1.68	54.77 ± 1.40	OOM
GTN [18]	93.52 ± 0.55	93.97 ± 0.54	60.47 ± 0.98	65.14 ± 0.45	91.31 ± 0.70	91.20 ± 0.71	OOM	OOM	OOM
RSHN [7]	93.34 ± 0.58	93.81 ± 0.55	59.85 ± 3.21	64.22 ± 1.03	90.50 ± 1.51	90.32 ± 1.54	OOM	OOM	OOM
HetGNN [43]	91.76 ± 0.43	92.33 ± 0.41	48.25 ± 0.67	51.16 ± 0.65	85.91 ± 0.25	86.05 ± 0.25	OOM	OOM	OOM
MAGNN [14]	93.28 ± 0.51	93.76 ± 0.45	56.49 ± 3.20	64.67 ± 1.67	90.88 ± 0.64	90.77 ± 0.65	OOM	OOM	OOM
HetSANN [8]	78.55 ± 2.42	80.56 ± 1.50	49.47 ± 1.21	57.68 ± 0.44	90.02 ± 0.35	89.91 ± 0.37	OOM	OOM	OOM
GCN [44]	90.84 ± 0.32	91.47 ± 0.34	57.88 ± 1.18	64.82 ± 0.64	92.17 ± 0.24	92.12 ± 0.23	27.84 ± 3.13	60.23 ± 0.92	OOM
GAT [45]	93.83 ± 0.27	93.39 ± 0.30	58.94 ± 1.35	64.86 ± 0.43	92.26 ± 0.94	92.19 ± 0.93	40.74 ± 2.58	65.26 ± 0.80	OOM
Simple-HGN [10]	94.01 ± 0.24	94.46 ± 0.22	63.53 ± 1.36	67.36 ± 0.57	93.42 ± 0.44	93.35 ± 0.45	47.72 ± 1.48	66.29 ± 0.45	OOM
HGT [9]	93.01 ± 0.23	93.49 ± 0.25	63.00 ± 1.19	67.20 ± 0.57	91.12 ± 0.76	91.00 ± 0.76	29.28 ± 2.52	60.51 ± 1.16	46.78 ± 0.42
GraphMSE [20]	94.08 ± 0.14	94.44 ± 0.13	57.60 ± 2.13	62.37 ± 1.03	92.58 ± 0.50	92.54 ± 0.14	OOM	OOM	OOM
DiffMG [22]	94.01 ± 0.37	94.20 ± 0.36	58.09 ± 1.35	59.75 ± 1.23	88.16 ± 2.83	88.07 ± 3.04	OOM	OOM	OOM
Random	93.57 ± 0.64	93.84 ± 0.53	52.13 ± 0.74	53.83 ± 0.66	90.91 ± 1.02	90.82 ± 0.93	21.22 ± 2.58	37.54 ± 2.66	35.14 ± 3.78
NARS [47]	94.18 ± 0.47	94.61 ± 0.42	63.51 ± 0.68	66.18 ± 0.70	93.36 ± 0.32	93.31 ± 0.33	49.98 ± 1.77	63.26 ± 1.26	50.66 ± 0.22
space4HGNN [48]	94.24 ± 0.42	94.63 ± 0.40	61.57 ± 1.19	63.96 ± 0.43	92.50 ± 0.14	92.38 ± 0.10	41.37 ± 4.49	65.66 ± 4.94	OOM
PMMM [23]	94.82 ± 0.26	95.14 ± 0.22	65.81 ± 0.29	67.58 ± 0.22	93.78 ± 0.25	93.71 ± 0.17	OOM	OOM	OOM
HINormer [11]	94.57 ± 0.23	94.94 ± 0.21	64.65 ± 0.53	67.83 ± 0.34	93.91 ± 0.42	93.83 ± 0.45	52.18 ± 0.39	64.92 ± 0.43	OOM
SeHGNN [16]	94.86 ± 0.14	95.24 ± 0.13	66.63 ± 0.34	68.21 ± 0.32	93.95 ± 0.48	93.87 ± 0.50	50.71 ± 0.44	63.41 ± 0.47	51.45 ± 0.29
SlotGAT [46]	94.95 ± 0.20	95.31 ± 0.19	64.05 ± 0.60	68.64 ± 0.33	93.99 ± 0.23	94.06 ± 0.22	49.68 ± 1.97	66.83 ± 0.30	OOM
LMSPS(ours)	95.35 ± 0.22	95.66 ± 0.20	66.99 ± 0.32	68.70 ± 0.26	94.73 ± 0.41	94.69 ± 0.36	53.26 ± 0.47	66.09 ± 0.51	54.83 ± 0.20
LPSMS(ours)	**95.77 ± 0.26**	**96.02 ± 0.17**	**68.26 ± 0.24**	**69.72 ± 0.32**	**94.98 ± 0.33**	**94.92 ± 0.25**	**53.87 ± 0.39**	**66.85 ± 0.36**	**55.17 ± 0.26**

OGBN-MAG is a large-scale dataset whose number of nodes is 10–175× larger than that of the other datasets.

complexities, indicating that performance gains are not tied to specific architectures or datasets. Instead, the results suggest that explicitly selecting informative relations is more effective than aggregating all available neighbors, which often introduces irrelevant or noisy signals.

Marginal contribution leads to more accurate relation estimation. LPSMS consistently achieves better performance than LMSPS, despite sharing the same backbone architecture. Since the only difference lies in the search strategy, this improvement can be attributed to more accurate estimation of relational importance. In particular, marginal contribution estimation isolates the effect of individual meta-paths, leading to more reliable subset selection compared to joint sampling strategies.

Performance gains are not due to model capacity. The MLP baseline performs significantly worse than HGNN-based methods, indicating that performance improvements do not originate from increased model complexity. Moreover, since our approach relies on a relatively simple MLP backbone during search, the gains are primarily driven by relational modeling rather than architectural enhancements such as attention or deep stacking.

Naive enumeration of relations is ineffective. Randomly selected meta-paths perform substantially worse than the proposed methods. This observation confirms that heterogeneous graphs contain a large number of redundant or even harmful relations, and that accurate identification of informative subsets is critical for effective learning.

Overall, these results provide strong empirical evidence that heterogeneous representation learning is fundamentally governed by relation selection rather than neighborhood expansion. By filtering out redundant relations and focusing on informative relational structures, the proposed framework achieves consistent improvements in both accuracy and scalability.

Table 4
Experiment results on the OGBN-MAG compared with other methods on the OGB leaderboard, using "LP" for label propagation, "MS" for multi-stage training, and "Emb" for extra embeddings.

Method	Trick	OGBN-MAG	
		Val Acc.	Test Acc.
SAGN [49]	+LP	52.25 ± 0.30	51.17 ± 0.32
GAMLP [50]	+LP	53.23 ± 0.23	51.63 ± 0.22
SeHGNN [16]	+LP	55.95 ± 0.11	53.99 ± 0.18
PSHGCN [51]	+LP	56.16 ± 0.21	54.57 ± 0.16
HGAMLP [52]	+LP	56.32 ± 0.06	54.61 ± 0.10
LMSPS	+LP	56.98 ± 0.10	55.10 ± 0.11
LPSMS	+LP	**57.15 ± 0.21**	**55.49 ± 0.19**
SAGN	+LP+MS+Emb	55.91 ± 0.17	54.40 ± 0.15
GAMLP	+LP+MS+Emb	57.02 ± 0.41	55.90 ± 0.27
SeHGNN	+LP+MS+Emb	59.17 ± 0.09	57.19 ± 0.12
PSHGCN	+LP+MS+Emb	59.43 ± 0.15	57.52 ± 0.11
HGAMLP	+LP+MS+Emb	59.60 ± 0.11	57.60 ± 0.10
LMSPS	+LP+MS+Emb	59.51 ± 0.07	57.84 ± 0.22
LPSMS	+LP+MS+Emb	**59.69 ± 0.25**	**57.96 ± 0.17**

6.4. Scalability on OGBN-MAG

We further verify scalability on million-scale graphs in Table 4.

With +LP. LPSMS achieves the best accuracy, showing that relational subset selection remains effective even under strong optimization tricks.

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Table 5
Experiments on OGBN-MAG to analyze the performance of SeHGNN, LMSPS and LPSMS under different maximum hops. #MP is the number of meta-paths under different maximum hops.

Max hop	#MP	SeHGNN		LMSPS		LPSMS	
		Time	Test accuracy	Time	Test accuracy	Time	Test accuracy
1	4	4.35	47.18 ± 0.28	3.98	46.88 ± 0.10	3.91	47.03 ± 0.14
2	10	6.44	51.79 ± 0.24	5.63	51.91 ± 0.13	5.88	52.14 ± 0.16
3	23	11.28	52.44 ± 0.16	10.02	52.72 ± 0.24	9.79	53.03 ± 0.21
4	50	OOM	OOM	14.34	53.43 ± 0.18	14.62	53.76 ± 0.20
5	107	OOM	OOM	14.77	53.90 ± 0.19	14.53	54.14 ± 0.15
6	226	OOM	OOM	14.71	54.83 ± 0.20	14.66	$\mathbf{55.17 \pm 0.26}$

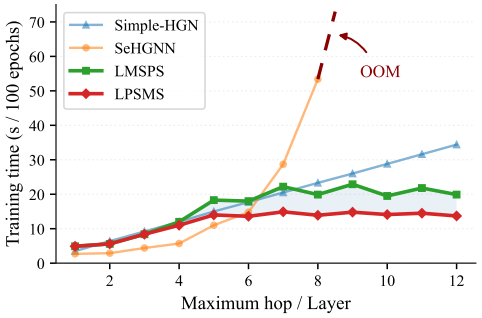


Figure 5: Illustration of average training time of Simple-HGN, SeHGNN, LMSPS(including search time), and LPSMS (including search time) relative to the maximum hop or layer on DBLP.

With +LP+MS+Emb. Performance gains remain consistent, indicating that improvements stem from relation estimation rather than training heuristics.

Discussion. Stable improvements under multiple training protocols demonstrate that the discovered relations represent intrinsic graph structure rather than overfitting.

6.5. Efficiency–Accuracy Trade-off

We analyze whether long-range dependency becomes effectively utilized only after relation selection in Figure 5.

Training cost vs hop. Traditional HGNNs suffer exponential growth or OOM, while our methods remain stable, verifying complexity independence from hop size.

Accuracy vs hop. Baseline models degrade at large hops due to noise accumulation. Our models remain stable and improve, demonstrating that long-range information is beneficial only after removing irrelevant relations.

Overall trade-off. Our methods lie on the Pareto frontier of accuracy and efficiency. LPSMS further improves accuracy without increasing cost, indicating better relational subset estimation rather than larger models.

Overall, these experiments show that long-range heterogeneous learning becomes feasible only after transforming aggregation into relational subset selection.

6.6. Effectiveness of Large Maximum Hops

We investigate how increasing the maximum hop affects learning on large-scale heterogeneous graphs. The goal of this experiment is not merely to test scalability, but to verify the central hypothesis of this paper: *performance improvement should come from selecting informative relations rather than expanding receptive fields.*

Table 5 reports the number of candidate meta-paths (#MP), training time, and accuracy on OGBN-MAG.

Exponential candidate growth vs. stable computation. As the maximum hop increases from 1 to 6, the number of candidate meta-paths grows exponentially from 4 to 226. This rapidly enlarges the combinational search space and makes conventional HGNNs impractical. SeHGNN already fails at hop 4 due to out-of-memory errors, demonstrating that directly expanding receptive fields in heterogeneous graphs is computationally prohibitive.

In contrast, both LMSPS and LPSMS remain stable. Although training time slightly increases from hop 1 to hop 3, it saturates once the candidate space exceeds the predefined subset size ($M = 30$). Even when #MP grows from 50 to 226 (hop 4 to hop 6), the per-epoch cost remains nearly unchanged. This confirms that the proposed progressive search converts exponential structural growth into constant computation.

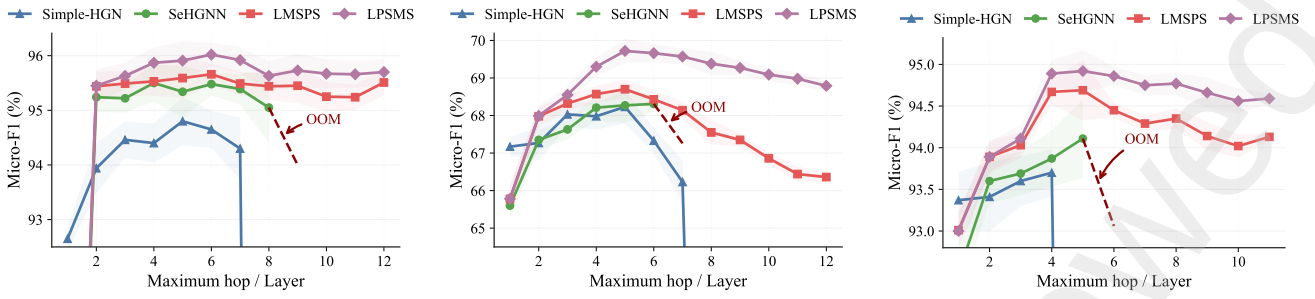
Accuracy improvement under controlled computation. Unlike traditional HGNNs where deeper propagation often harms performance, accuracy consistently improves with hop for both methods. The performance reaches 54.83% for LMSPS and **55.17**% for LPSMS at hop 6.

This behavior indicates that long-range dependencies are indeed useful, but only after removing redundant relations. Increasing hop enlarges the candidate evidence pool, while the search algorithm filters useful signals from it. Therefore performance grows without introducing noise.

Why single-path search helps. Under identical computational budgets, LPSMS consistently outperforms LMSPS. This suggests that estimating marginal contribution (leave-one-out) produces more reliable relation importance than joint sampling. Consequently, the model benefits more from better selection rather than deeper propagation.

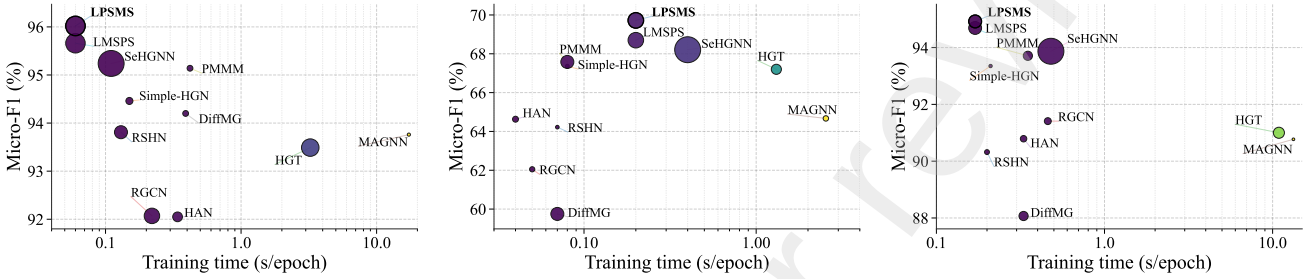
Conclusion. Naively increasing hop causes exponential explosion and memory failure, while progressive selection

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(a) DBLP (b) IMDB (c) ACM

Figure 6: Performance versus maximum hop/layer: large-hop robustness and OOM behavior.



(a) DBLP (b) IMDB (c) ACM

Figure 7: Pareto comparison of HGNNs in accuracy, training efficiency, and model complexity.

maintains constant cost and improves performance. This verifies that long-range dependency in heterogeneous graphs is limited by relational redundancy, not propagation distance.

6.7. Effectiveness of the Search Algorithm

To isolate the effect of the search mechanism, we replace our search strategy with representative differentiable NAS algorithms, including DARTS, SPOS, DiffMG, and PMMM, while keeping the same backbone and derivation procedure. Figure 8 reports Micro-F1 on four datasets.

LMSPS consistently outperforms all alternative search strategies. For instance, on DBLP it achieves 95.66%, exceeding the strongest competitor PMMM (95.48%). Similar margins appear on IMDB (68.70% vs. 67.49%), ACM (94.69% vs. 93.74%), and Freebase (66.09% vs. 64.83%).

LPSMS further improves results across all datasets (95.89%, 69.72%, 94.92%, 66.37% respectively), showing that marginal contribution estimation is more reliable than joint sampling.

These results demonstrate that performance gains originate from identifying informative relations, not from architectural complexity. The search algorithm itself determines effectiveness.

6.8. Effectiveness of the Searched Meta-Paths

We next evaluate whether the discovered meta-paths generalize across architectures. If the paths capture dataset semantics rather than model bias, they should improve other HGNNs.

We transfer the discovered meta-paths to HAN and SeHGNN, keeping all other settings unchanged.

Figure 9 shows consistent improvements across datasets.

On HAN, LMSPS improves Freebase from 54.77% to 57.13%, and LPSMS further to 57.91%. Similar gains appear on DBLP (92.05% → 93.54% → 93.77%), IMDB (64.63% → 65.89% → 66.02%), and ACM (90.79% → 92.28% → 92.69%).

Even on the strong SeHGNN backbone, performance still improves consistently. This indicates the discovered relations reflect intrinsic dataset structure.

Overall, the transferability confirms that the framework learns reusable relational patterns rather than model-specific heuristics.

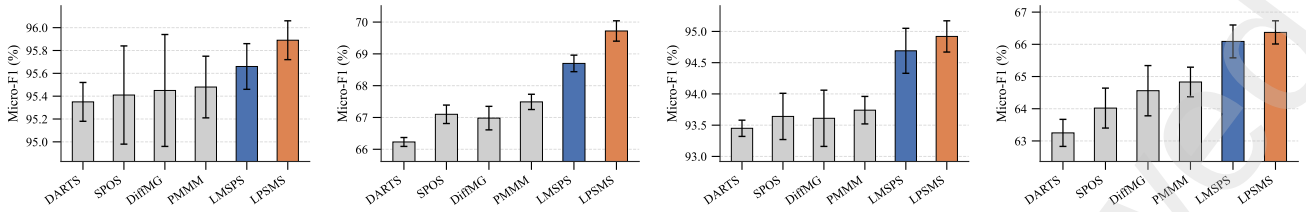
7. Interpretability and Structural Analysis

Previous sections demonstrate that the proposed framework improves accuracy, scalability, and robustness. In this section, we provide an interpretability analysis to understand *what relational structures the model learns and why they improve performance*.

Our central claim is that the model does not primarily benefit from enlarging receptive fields, but from identifying a stable subset of informative relations. In heterogeneous graphs, long-range propagation alone amplifies both useful signals and irrelevant semantics. The key factor is therefore whether the model can distinguish structural evidence from relational noise.

To validate this hypothesis, we analyze whether the discovered meta-paths correspond to intrinsic relational structure rather than optimization artifacts. This analysis provides direct interpretability by linking selected meta-paths to meaningful semantic relations in each dataset.

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(a) DBLP (b) IMDB (c) ACM (d) Freebase

Figure 8: Effectiveness of different meta-Path search algorithms.

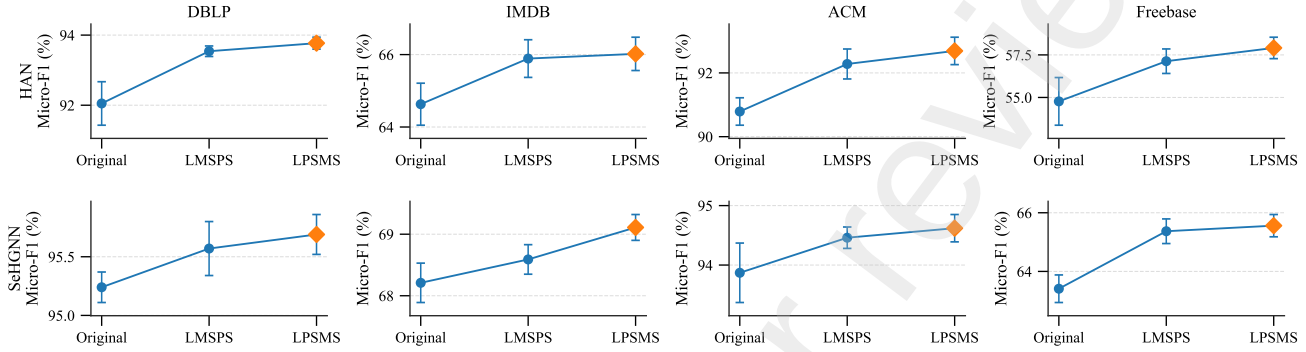


Figure 9: Transferability of the searched meta-paths across different HGNNs.

7.1. Interpretable Relational Structures

Tables 6 and 7 group the discovered meta-paths by hop length.

Unlike conventional HGNNs that rely on short manually designed paths, our framework automatically selects paths ranging from 1-hop to 6-hop. Importantly, longer paths do not introduce unrelated semantics; instead, they refine shorter relations hierarchically.

For example, on OGBN-MAG, the 1-hop meta-path PF encodes topical relation, while higher-hop paths such as PAPAPF and PFPAPF integrate citation and collaboration context. These longer paths do not replace short relations but provide corroborating evidence. The monotonic improvement observed in Table 5 indicates that long-range dependencies become useful only after filtering irrelevant relations.

Table 6
Meta-paths searched by LMSPS grouped by hop number.

Hop	DBLP	IMDB	ACM	OGBN-MAG
0-hop	–	M	–	–
1-hop	AP	MA, MK	–	PF
2-hop	APT	MAM, MDM, MKM	PAP	–
3-hop	APVP	MAMK, MDMK, MKMD	PAPP, PCPA, PCPP, PPPC, PPPP	PAPF, PFPP
4-hop	APAPA, APTPA, APTPT, APTPV, APVPA, APVPV	MDMKM, MKMAM, MKMDM, MKMKM	PAPAP, PAPPP, PCPAP, PCPPA, PPAPA, PPAPC, PPAPP	PAPPP, PFPFP, PPAPP, PPPAP, PPPFP
5-hop	APAPAP, APAPTP, APAPVP, APTPTP, APTPVP, APVPTP	MAMAMK, MAMDMA, MAMDMK, MAMKMD, MAMKMK, MDMAMA, MDMAMD, MDMDMA, MDMDMK, MDMKMA, MKMAMA, MKMAMD, MKMAMK, MKMDMA, MKMDMK, MKMKMD, MKMKMK	PAPAPA, PAPCPA, PAPPAP, PAPPCP, PAPPPP, PCPAPP, PCPCPP, PCPPAP, PCPPPP, PPAPAP, PPAPCP, PPAPPA, PPAPPP, PPCPAP, PPPAPA, PPPCPA, PPPPPP	PAIAPP, PAPAPF, PFPAPF, PFPPPF, PFPPPP, PPAPPF, PPPAPF, PPPPAP, PPPPPF
6-hop	APAPAPV, APAPTPA, APAPTPV, APAPVPA, APAPVPT, APAPVPV, APTPAPA, APTPAPT, APTPTPT, APTPVPV, APVPAPT, APVPTPA, APVPVPA, APVPVPT, APVPVPV	–	–	PAIAPAP, PAPAPAP, PAPAPPP, PAPPPAI, PAPPPPF, PAPPPPP, PFPAPAP, PFPPFPF, PFPPPPF, PPAPAPF, PPAPAPP, PPPAIAI, PPPAPPP

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Table 7
Meta-paths searched by LPSMS grouped by hop number.

Hop	DBLP	IMDB	ACM	OGBN-MAG
0-hop	–	M	–	–
1-hop	AP	MA, MD, MK	PA, PC	PA, PI, PF
2-hop	APA, APT, APV	MAM, MDM, MKM	PAP, PCP	PAP, PIP, PFP
3-hop	APAP, APTP, APVP	MAMA, MDMD, MKMK, MAMD	PAPA, PCPC, PAPC	PAPA, PIPI, PFPF, PAPF
4-hop	APAPA, APAPT, APAPV, APTPA, APTPT	MAMAM, MAMDM, MAMKM, MDMDM, MKMKM	PAPAP, PAPCP, PCPAP, PCPCP, PPACP	PAPAP, PIPIP, PFPFP, PAPFP, PFPAP
5-hop	APAPAP, APAPTP, APAPVP, APTPAP, APTPTP	MAMAMA, MAMDMA, MAMKMK, MDMAMA, MKMKMA, MAMDMK, MDMDMA, MKMAMK, MDMKMK, MKMDMK, MAMKMA, MDMAMK, MKMAMD, MDMAMM	PAPAPA, PAPAPC, PAPCPA, PAPCPC, PCPAPA, PCPAPC, PCPCPA, PCPCPC, PAPCAP, PCPACP, PAPCPC, PAPCPP, PCPAPP, PAPACP, PCPCAP, PAPAPC, PCPAPA, PAPCPA	PAPAPA, PIPIPI, PFPFPF, PAPFPA, PFPAPA
6-hop	APAPAPA, APAPAPT, APAPAPV, APAPTPA, APAPTPV, APAPVPA, APTPAPA, APTPAPT, APVPAPA, APAPVPV, APTPVPA, APVPTPA, APVPVPA	–	–	PAPAPAP, PIPIPIP, PFPFPFP, PAPIPAP, PAPFPAP, PFPAPFP, PIPAPIP, PAPIPFP, PFPIPAP, PAPFPFP

This suggests that the limitation of heterogeneous learning is not propagation depth itself, but the inability to separate informative relational channels from combinational noise. After relation selection, deeper aggregation becomes beneficial rather than harmful, supporting our formulation that heterogeneous learning is fundamentally a selection problem rather than a propagation problem.

Dataset-dependent semantics. The selected node types align closely with prediction objectives.

On OGBN-MAG, most paths originate from Paper and Field, indicating venue prediction depends primarily on citation context and research topics. On DBLP, Author–Paper relations dominate, reflecting collaboration-driven labeling. On IMDB, Movie–Actor–Director relations appear frequently, capturing production semantics. On ACM, Paper–Conference relations are emphasized.

This consistency across datasets shows that the discovered meta-paths reflect dataset-level semantics rather than architecture-specific biases. This demonstrates that the model captures semantically meaningful and task-relevant relational structures, rather than relying on spurious correlations.

7.2. Marginal Contribution and Structural Stability

Comparing LMSPS and LPSMS reveals a qualitative difference: LPSMS discovers more regular structures such as APAPA and PFPFPF, while LMSPS produces more heterogeneous combinations. This difference reflects the optimization objective.

Multi-path sampling estimates joint compatibility among relations, favoring combinations that co-adapt during training. In contrast, leave-one-out evaluation estimates marginal contribution, favoring relations that remain useful regardless of surrounding context.

Marginal contribution therefore identifies structurally stable relations, whereas joint optimization may overfit co-occurrence patterns. This explains the improved transferability and robustness observed in previous sections.

Consequently, LPSMS is not merely a faster variant: it changes the learning objective from subset fitting to structure estimation. This also enhances interpretability, as each selected meta-path can be directly attributed to its marginal contribution.

7.3. Component-wise Estimation of Relational Importance

As shown in Table 8, each module in the framework can be unified as performing relational importance estimation.

Progressive Sampling. Removing this component exposes the full combinatorial search space and leads to substantial performance degradation. This suggests that failures at large hop distances are primarily caused by irrelevant relations rather than excessive propagation depth.

Sampling Evaluation. Without explicit evaluation, relation importance becomes noisy and unreliable. This indicates that simple statistics such as frequency or magnitude are insufficient to characterize relational utility.

Single-path Exploration. Simultaneous optimization over multiple relations introduces gradient interference. Restricting exploration to a single path stabilizes the estimation process and reduces coupling among candidate relations.

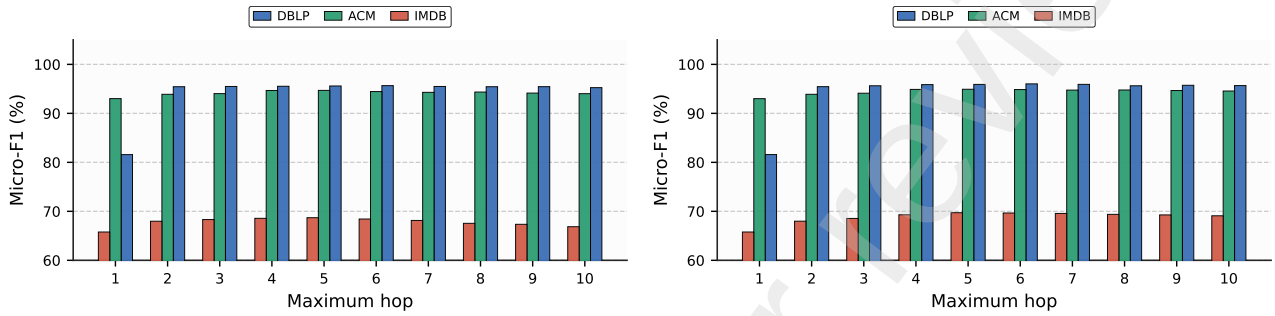
Leave-One-Out. Explicit marginal contribution estimation further improves performance. This demonstrates that informative relations are those whose utility remains stable under contextual perturbation.

Taken together, these components form a coherent mechanism for relational subset estimation, explaining the increasing performance gains observed as graph scale grows. From an interpretability perspective, each module progressively isolates and validates the contribution of individual relations.

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Table 8
Experiments on small and large datasets to analyze the effects of different blocks in LMSPS and LPSMS. *PS* means progressive sampling strategy, *SE* means sampling evaluation strategy, *SP* means progressive single meta-path search algorithm, and *LLO* means leave-one-out strategy.

Method	DBLP		IMDB		ACM		Freebase		OGBN-MAG
	Macro-F1	Micro-F1	Macro-F1	Micro-F1	Macro-F1	Micro-F1	Macro-F1	Micro-F1	Test Acc.
LMSPS w/o *PS*	94.71 ± 0.23	95.00 ± 0.19	64.85 ± 0.46	66.52 ± 0.37	93.19 ± 0.34	93.14 ± 0.41	48.89 ± 0.47	61.61 ± 0.51	47.66 ± 0.45
LMSPS w/o *SE*	95.15 ± 0.28	95.48 ± 0.24	65.46 ± 0.48	67.13 ± 0.47	94.20 ± 0.35	94.15 ± 0.31	52.08 ± 0.33	64.84 ± 0.38	52.94 ± 0.34
LMSPS	$\mathbf{95.35 \pm 0.22}$	$\mathbf{95.66 \pm 0.20}$	$\mathbf{66.99 \pm 0.32}$	$\mathbf{68.70 \pm 0.26}$	$\mathbf{94.73 \pm 0.41}$	$\mathbf{94.69 \pm 0.36}$	$\mathbf{53.26 \pm 0.47}$	$\mathbf{66.09 \pm 0.51}$	$\mathbf{54.83 \pm 0.20}$
LPSMS w/o *SP*	95.11 ± 0.21	95.36 ± 0.23	66.74 ± 0.35	68.58 ± 0.26	93.55 ± 0.26	93.78 ± 0.35	50.96 ± 0.47	64.61 ± 0.43	49.89 ± 0.34
LPSMS w/o *LOO*	94.45 ± 0.26	94.78 ± 0.33	66.81 ± 0.32	67.44 ± 0.32	94.11 ± 0.24	94.06 ± 0.23	52.17 ± 0.24	64.96 ± 0.34	53.22 ± 0.36
LPSMS	$\mathbf{95.77 \pm 0.26}$	$\mathbf{96.02 \pm 0.17}$	$\mathbf{68.26 \pm 0.24}$	$\mathbf{69.72 \pm 0.32}$	$\mathbf{94.98 \pm 0.33}$	$\mathbf{94.92 \pm 0.25}$	$\mathbf{53.87 \pm 0.39}$	$\mathbf{66.85 \pm 0.36}$	$\mathbf{55.17 \pm 0.26}$



(a) LMSPS (b) LPSMS

Figure 10: Micro-F1 scores with respect to different maximum hops on DBLP, IMDB, and ACM .

7.4. Long-Range Stability

We analyze the sensitivity to maximum hop in Figure 10.

Performance improves and then stabilizes, rather than degrading with increasing depth, in contrast to conventional HGNNs. This suggests that instability in heterogeneous graphs is primarily driven by relational redundancy rather than propagation distance.

After redundant relations are filtered out, long-range dependencies become consistently informative instead of introducing noise.

Overall, the results indicate that the effectiveness of the framework does not stem from deeper aggregation, but from identifying stable and informative relational structures within heterogeneous graphs.

8. Conclusion

In this work, we revisit heterogeneous graph representation learning from a relational perspective. Instead of attributing the limitations of HGNNs to insufficient propagation depth, we show that the core challenge lies in identifying informative relations from an exponentially large candidate space.

Based on this insight, we formulate heterogeneous representation learning as a relational subset selection problem, which shifts the focus from expanding receptive fields to selecting informative relational structures. We propose a progressive meta-path search framework that incrementally reduces the candidate space while preserving useful long-range dependencies. In addition, we introduce a progressive single-path search strategy based on marginal contribution estimation, which improves both efficiency and robustness by approximating combinatorial optimization with tractable evaluation.

Extensive experiments demonstrate that the proposed methods consistently outperform existing HGNNs on both small-scale and large-scale datasets. More importantly, the discovered meta-paths generalize across different model architectures, indicating that the framework captures dataset-level relational patterns rather than model-specific behaviors.

Overall, our results suggest that effective heterogeneous representation learning depends less on deeper propagation and more on identifying stable and informative relational structures. This perspective provides a principled direction for designing scalable and interpretable graph learning methods, and may inspire future research beyond architecture-centric approaches.

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